

Topic 3: Basic Prompt Structure and Formatting

3.1 Anatomy of a Good Prompt

- Role: who the model should act as (optional but often helpful)
- Task: what you want the model to do -- be specific and use action verbs
- Context: background information the model needs to complete the task
- Input data: the actual content to process (article to summarise, code to review, etc.)
- Output format: how you want the answer structured (bullet list, JSON, paragraph, table)
- Constraints: length limits, tone, language, things to avoid

```
Role:    You are a senior software engineer.
Task:    Review the following Python function for bugs and
         performance issues.
Context: This function is used in a high-traffic web API.
Input:   [paste function here]
Format:  Return a numbered list of issues, each with:
         - Issue description
         - Severity (Low / Medium / High)
         - Suggested fix
Avoid:   Do not rewrite the entire function.
```

3.2 Being Specific vs Being Vague

- Vague: 'Tell me about climate change'
- Specific: 'Summarise the three main causes of accelerated Arctic ice melt since 2000, in 150 words, for a high-school audience'
- Vague: 'Write me some code'
- Specific: 'Write a Python function that takes a list of integers and returns the top 3 unique values sorted descending. Include type hints and a docstring.'
- Rule of thumb: if a human colleague would need to ask you clarifying questions, your prompt is too vague

3.3 Using Delimiters to Structure Input

- Delimiters separate instructions from data, preventing prompt injection and confusion
- Common delimiters: triple backticks (``), XML tags (<text></text>), triple quotes (---)
- Always tell the model what each delimited block contains

```
Summarise the article below in 3 bullet points.
```

```
Article:
```

```
```
```

```
Scientists at MIT have developed a new battery technology...
```

```
```
```

```
OR using XML tags:
```

```
<article>
```

```
Scientists at MIT have developed a new battery technology...
```

```
</article>
```

```
Summarise the article above in 3 bullet points.
```

3.4 Formatting Output

- Ask explicitly for the format you need: 'Return your answer as a JSON object', 'Use a markdown table', 'Respond in bullet points'
- Provide a template or example of the desired output format for highest accuracy
- For structured data (JSON, CSV) always validate the output -- models can make small formatting errors
- Use 'Do not include any explanation, just the JSON' to suppress unwanted prose

```
Extract the following fields from the job posting below.
```

```
Return ONLY a JSON object with no additional text.
```

```
Fields: job_title, company, location, salary_range, required_skills (array)
```

```
Job posting:
```

```
```
```

```
[job posting text here]
```

```
```
```